

Group-22

Zero-Shot Object Counting with Language-Vision Models

DLP 2025 Summer Paper Presentation

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Introduction

Background & Motivation

- Object counting is important in many applications

e.g. crowd counting, traffic monitoring, wildlife analysis

- Existing methods require labeled exemplars for each test class
- This is costly and not scalable → motivates zero-shot object counting

Challenges

- No human-annotated exemplars in the zero-shot setting
- Variations in object appearance (pose, scale, background)
- Need to connect language supervision with vision models
- How to select good exemplars without ground-truth labels?

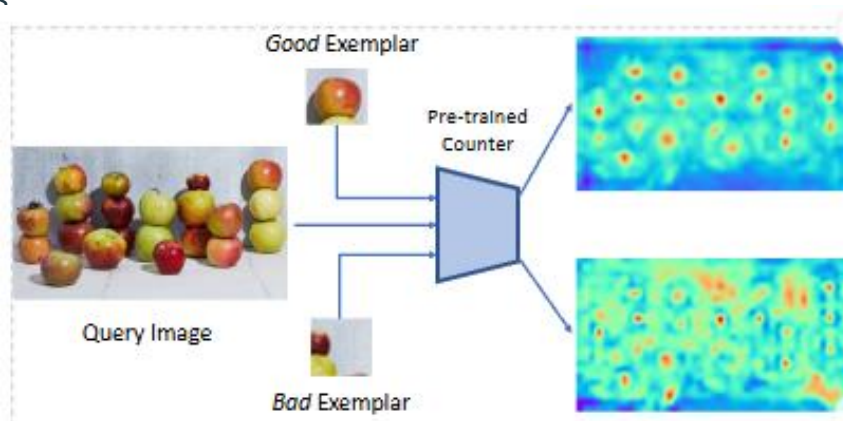
Contributions (from the paper)

- Introduce Zero-Shot Object Counting framework
- Propose two ways to construct class prototypes (VAE-based, SD-based)
- Develop an error predictor to select best exemplars automatically
- Achieve strong performance without any manual exemplar labeling

Related Work

Exemplar-based Object Counting

- Traditional methods: require exemplar patches at test time
- Key limitation: need human-provided exemplars
- Cannot generalize to unseen categories



Zero-shot / Few-shot Learning

- Zero-shot learning: transfer knowledge from language supervision
- Few-shot object counting: use very few exemplars but still needs labels
- Our method: fully zero-shot, no manual labeling, only class name

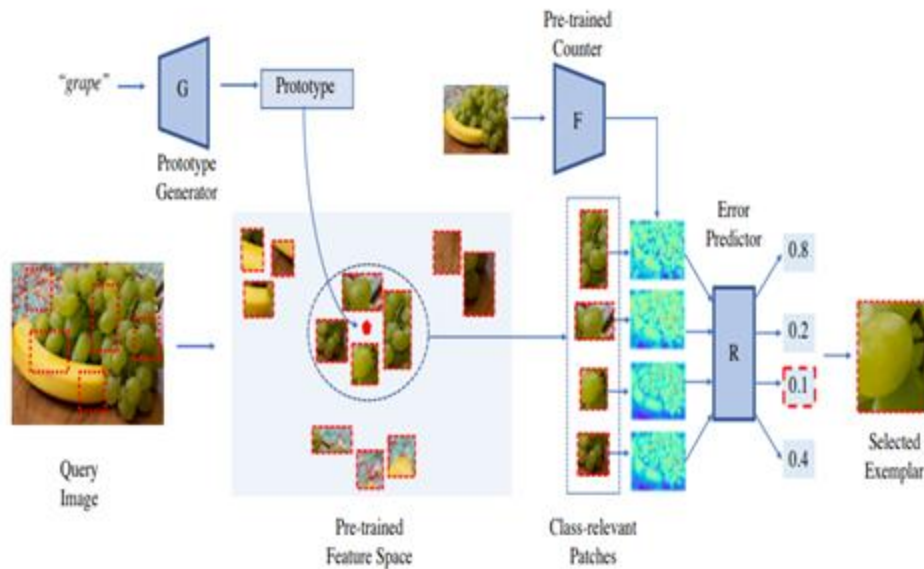
Vision-Language Models

- CLIP, Stable Diffusion, and large-scale pretrained models
- Enable bridging between class names and visual representations
- Our approach leverages these models to construct class prototypes

Method

Overview

- **Base Counting Model**
- **Zero-shot Counting**
 1. Class-relevant Patch Selection
 2. Exemplar Selection (Error Predictor)
 3. Final Counting



Base Counting Model

- **Input:** Query image I + exemplar patch E
- **Process:**
 1. Extract features $F(I), F(E)$
 2. Compute similarity map S
 3. Counter module $\rightarrow$ Density map $D \rightarrow$ Predicted count
- **Loss:** $L_{\text{count}} = \|D(I, B) - D^*(I)\|_2^2$

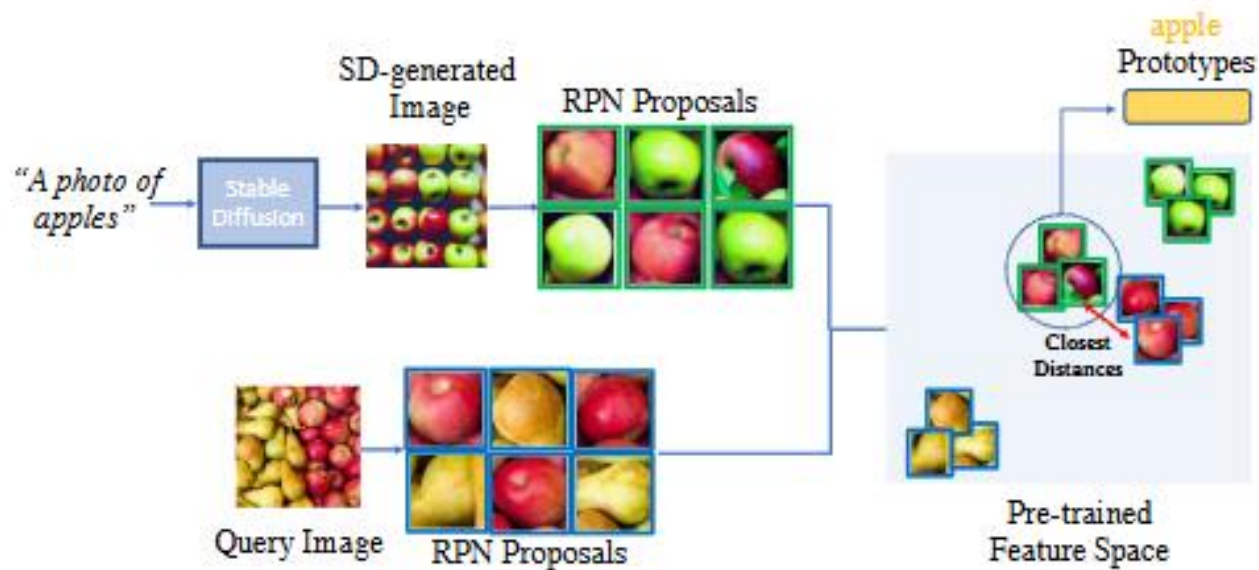
L2 loss between predicted and ground truth density maps

Zero-shot Counting – Step 1

Selecting Class-relevant Patches:

1. Construct class prototype:
 - ❑ **VAE-based:** Generate features from semantic embeddings
 - ❑ **Stable Diffusion-based:** Generate synthetic images → crop patches → aggregate features
1. Extract candidate patches using RPN
2. Match candidate patches with class prototype → select class-relevant patches

Zero-shot Counting – Step 1



Zero-shot Counting – Step 2

Selecting Exemplars for Counting:

- Problem: Not all class-relevant patches are suitable as exemplars
 - Solution: Introduce an **Error Predictor** R
- Predicts the counting error if a patch is used as exemplar

$$\epsilon = \left| \sum_{i,j} \bar{D}_{(i,j)} - \bar{N}^* \right|$$

Error Predictor Training

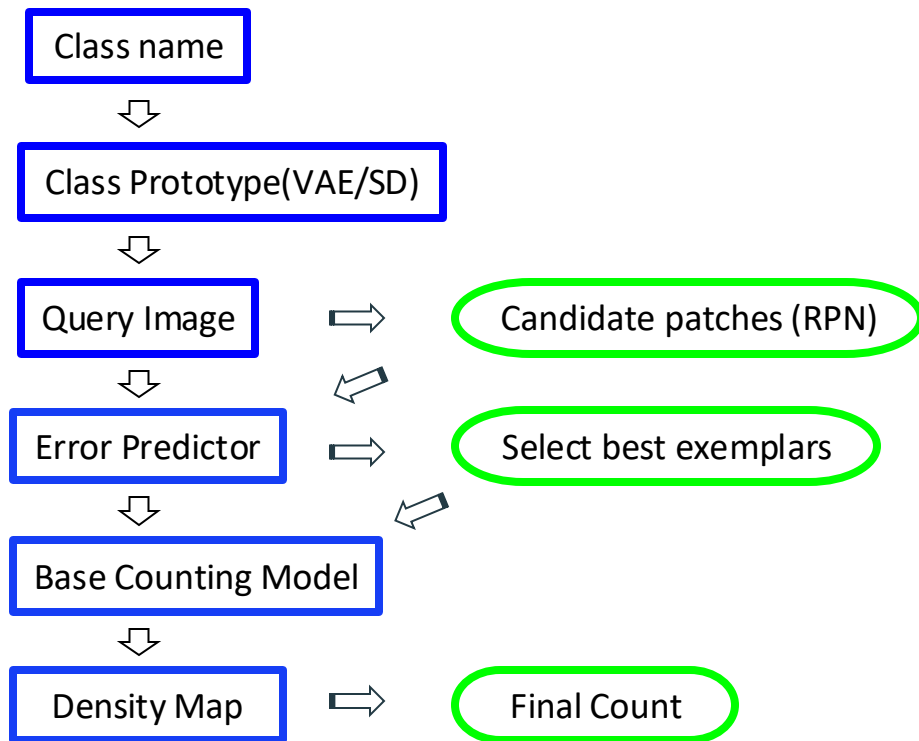
- Input:
 1. Image feature map $F(I)$
 2. Similarity map S
- Output: Mean Squared Error between predicted error and actual error
- Select patches with lowest predicted error as exemplars

Zero-shot Counting – Step 3

Final Counting

- Use selected exemplar patches
- Feed them into the base counting model
- Output: Density map $\rightarrow$ Final object count

Method Summary



Experiment

Network Architecture

Base Counting Model

- Backbone: ResNet-50 (pre-trained on ImageNet)
- Feature Maps: 1024 $\rightarrow$ 256 channels (1 $\times$ 1 conv)
- Exemplar Processing: Global average pooling $\rightarrow$ 256-d vector
- Counter: 5 conv + bilinear upsampling layers

Feature Generation Model

- Encoder/Decoder: 2-layer FC networks (4096 hidden units)
- Activations: LeakyReLU (hidden) + ReLU (output)
- Latent Space: 512 dimensions
- Semantic Embeddings: 512 dimensions

Error Predictor

- Architecture: 5 conv + bilinear upsampling + linear layer
- Output: Counting error prediction

Experimental Setup

- Dataset: FSC-147 (Large-scale class-agnostic counting dataset)
 - 6,135 images across 147 categories
 - Non-overlapping categories in train/val/test splits
- Evaluation Metrics:
 - Mean Absolute Error (MAE)
 - RMSE (Root Mean Squared Error)
 - NAE (Normalized Absolute Error)
 - SRE (Squared Relative Error)

Main Results

Few-Shot vs Zero-Shot Counting on FSC-147

Type	Method	Test MAE	Test RMSE	Improvement
Few-Shot Methods (with human annotations)	FamNet+	22.08	99.54	Baseline
	BMNet+	14.62	91.83	Best few-shot
Their Zero-Shot Methods	VAE	23.14	114.40	-37% vs RPN
	Stable Diffusion	21.53	113.28	-42% vs RPN
Reference-less baseline	RPN only	29.25	130.65	-

Ablation Studies Component Analysis

- Prototype Generation Methods (in MAE): CLIP (30.32) > VAE (23.14) > SD-based (21.53)
- Error Predictor Impact (in MAE): Without (21.99) > With (21.53)
- Key Components Contribution
 - Class prototype selection: Reduces MAE by 4.87-7.26
 - Error predictor: Additional MAE reduction of 4.63-6.46
 - **Combined approach: Best performance**

Prototype	Predictor	Val Set				MAE	Test Set			
		MAE	RMSE	NAE	SRE		MAE	RMSE	NAE	SRE
-	-	32.19	99.21	0.38	4.80	29.25	130.65	0.35	4.35	
✓(VAE)	-	29.85	97.63	0.42	5.20	24.38	115.04	0.36	4.18	
✓(SD)	-	27.76	97.06	0.35	4.43	21.99	113.31	0.33	3.77	
-	✓	27.56	88.57	0.41	4.54	22.79	127.09	0.36	3.98	
✓(VAE)	✓	27.47	90.85	0.37	4.52	23.14	114.40	0.34	3.95	
✓(SD)	✓	26.30	88.80	0.34	4.27	21.53	113.28	0.31	3.61	

TABLE 2

Ablation study on each component's contribution to the final results. We show the effectiveness of the two steps of our framework: selecting class-relevant patches via a generated class prototype and selecting optimal patches via an error predictor.

Conclusions - Contributions & Results

- Main Contributions
 - Introduced zero-shot counting task
 - Leveraged language-vision models for prototype generation
 - Developed error-guided patch selection
- Results on FSC-147
 - Best Performance: 21.53 MAE (SD-based)
 - Improvement: 7.26 MAE reduction vs VAE
 - Generalization: Works with multiple counting models
- Impact
 - Eliminates need for manual annotations
 - Bridges language and vision for counting tasks

Thank you for listening